Simetría $\sigma_{ij}=\sigma_{ji}$

$$\frac{\partial \sigma_{ij}}{\partial x_j} + X_i = 0$$

$$V_i = \frac{\partial \sigma_{ji}}{\partial x_j} + X_i = 0 \qquad \forall x \in \Omega$$

fuerza másicas o de cuerpo $\equiv 0$

$$\sigma_x = 3x^2 + 4xy - 8y^2 \qquad \tau_{xy} = -\frac{1}{2}x^2 - 6xy - 2y^2$$

$$\sigma_y = 2x^2 + xy + 3y^2 \qquad \sigma_z = \tau_{xz} = \tau_{yz} = 0$$

$$\underline{\underline{\sigma}} = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_y & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_z \end{bmatrix} = \begin{bmatrix} \sigma_x & \tau_{xy} & 0 \\ \tau_{xy} & \sigma_y & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{cases} \dfrac{\partial \sigma_{xx}}{\partial x} + \dfrac{\partial \sigma_{yx}}{\partial y} + \cancel{\dfrac{\partial \sigma_{zx}}{\partial z}} + \cancel{X_x} = 0 \\ \dfrac{\partial \sigma_{xy}}{\partial x} + \dfrac{\partial \sigma_{yy}}{\partial y} + \cancel{\dfrac{\partial \sigma_{zy}}{\partial z}} + \cancel{X_y} = 0 \\ \cancel{\dfrac{\partial \sigma_{xz}}{\partial x}} + \cancel{\dfrac{\partial \sigma_{yz}}{\partial y}} + \cancel{\dfrac{\partial \sigma_{zz}}{\partial z}} + \cancel{X_z} = 0 \end{cases} \qquad \begin{cases} (1)\ \dfrac{\partial \sigma_x}{\partial x} + \dfrac{\partial \tau_{xy}}{\partial y} = 0 \\ (2)\ \dfrac{\partial \tau_{xy}}{\partial x} + \dfrac{\partial \sigma_y}{\partial y} = 0 \end{cases} \qquad \forall \underline{x} \in \Omega$$

$$\frac{\partial \sigma_x}{\partial x} = 6x + 4y \qquad \frac{\partial \tau_{xy}}{\partial x} = -x - 6y$$

$$\frac{\partial \sigma_y}{\partial y} = x + 6y \qquad \frac{\partial \tau_{xy}}{\partial y} = -6x - 4y$$

$$\left.\begin{array}{ll} (1) & \cancel{6x} + \cancel{4y} - \cancel{6x} - \cancel{4y} = 0 \quad \forall\, \underline{x} \in \Omega \\ (2) & -\cancel{x} - \cancel{6y} + \cancel{x} + \cancel{6y} = 0 \quad \forall\, \underline{x} \in \Omega \end{array}\right\} \Rightarrow$$

El cuerpo está en equilibrio